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Junyu Cao, Wei Qi

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Stall Economy: The Value of Mobility in Retail on Wheels

Junyu Cao,^a Wei Qi^{b,*}

^aMcCombs School of Business, The University of Texas at Austin, Austin, Texas 78712; ^bDepartment of Industrial Engineering, Tsinghua University, Beijing 100084, China

*Corresponding author

Contact: junyu.cao@mcombs.utexas.edu,  <https://orcid.org/0000-0001-9235-1411> (JC); qiw@tsinghua.edu.cn,

 <https://orcid.org/0000-0003-3948-835X> (WQ)

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Abstract. Urban open space emerges as a new territory to embrace retail innovations. Selling products in public spaces with wheeled stalls can potentially become ubiquitous in our future cities. Transition into such a “stall economy” paradigm is being spurred by the rapidly advancing self-driving technologies. Motivated by this transformation, this paper provides models, theory, and insights of spatial queueing systems, in which one server moves around to meet mobile customers/machines and in which the “last 100 meters” are expensive. Specifically, we study two service modes: (i) on-demand, first come, first served and (ii) spatially and temporally pooling customer demands. In each mode, we derive the dependence of customer waiting and stall repositioning on two key decisions: the service zone size and the walking distance imposed on customers to meet a stall. In particular, for the on-demand mode, we propose and solve a “rendezvous problem” to analytically characterize the spatial distribution of the stall-customer meeting locations. We also propose a stylized joint truck-stall routing model to capture the inventory replenishment operations. Our main finding is that the stall economy potentially profits more than stationary retail, not only because of the mobility of stalls for providing proximity to customers, but also because of its operational flexibilities that allow for avoiding the “last 100 meters” and pooling demands. In a broader sense, this work looks toward an expanded scope of future retail empowered by self-driving technologies.

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Keywords: stall economy • mobile retail • self-driving • rendezvous problem • spatial queues

1. Introduction

Shopping is a central aspect of urban life, but the ways of shopping are constantly changing. Moving forward to a smart-city future, cities are embracing the emergence of new retail paradigms. That is, retail channels diversify and blur the traditional offline and online boundaries, combining the best of both options. Intriguing newer forms of retail include buy online, pick up in store (e.g., Walmart); buy online, fulfill and deliver by a shopper (e.g., Instacart); and e-commerce with physical showrooms (e.g., ModCloth).

Despite the remarkable feats of these retail channel innovations, they have largely been confined to two territories: brick-and-mortar stores and the internet. Meanwhile, urban open spaces have largely been overlooked but now, have the potential to become a third territory for retail innovation. The advances in self-driving technologies create the prospect of a “stall economy”—the mode of selling

products in public spaces with wheeled vending stalls. As an example of practicing the stall economy, the Stop & Shop grocery chain allows its customers in the Boston area to simply tap a button on their smartphone to summon a “Robomart,” which is a self-driving mini grocery store (robomart.co). Customers also can browse for products on their smartphone. Once the mini store arrives nearby, the customer can go to the store, unlock the store’s doors, and select the groceries they would like to purchase (see Figure 1). Stop & Shop promoted this retail mode as follows (Stop & Shop 2019).

The vehicles will address consumers’ desires to select their own fresh produce when shopping via online or mobile and will deliver an even more convenient grocery shopping experience for Stop & Shop customers ... This is one way in which we’re leveraging new technology to make shopping easier for our customers—by essentially bringing the store to them.

Figure 1. (Color online) Shopping at a Self-Driving Stall



Source: TechSpot (2019).

Can this self-driving-enabled stall economy really prosper? This retail channel is still in its infancy, and its prospect is a bit unclear. On the optimistic side, the stall economy indeed has the following two advantages.

- **Mobility.** As the Stop & Shop quote boasts, dispatching mobile stalls to customers enhances customer utility by partially resolving the classical dilemma between online and offline channels. Compared with online retail, brick-and-mortar stores can engage customers in a touch and feel shopping experience and avoid the last-mile fulfillment cost and delayed delivery, but visiting stores also imposes a travel disutility on customers.

- **Asset lightness.** Purchasing or leasing mobile stalls frees retailers from binding themselves to real estate and thus, generally incurs less up-front investment and maintenance cost than purchasing or leasing traditional brick-and-mortar stores.

Moreover, self-driving vehicles are already gaining momentum in the arena of last-mile deliveries, with major players, such as Walmart (in partnership with Cruise) in the United States (Walmart 2021) and Meituan in China (Global Times 2021), piloting such projects. In addition, some start-ups, such as Nuro (nuro.ai) and Neolix (neolix.ai), are racing to launch unmanned delivery vehicles. These endeavors may pave the way for further engaging self-driving vehicles as mobile convenience stores on a large scale.

Meanwhile, on the pessimistic side, the extent to which we can extrapolate from the growth of delivery services is unclear. These delivery services fulfill online orders. In contrast, the stall economy itself is a retail channel and bears its own unique, intrinsic challenges (which come hand in hand with the two advantages).

- Although mobility allows a stall to reach more customers, offering close proximity by moving and perching at more locations than a brick-and-mortar store, doing so also entails vehicle operating costs and customer wait times. One way to reduce vehicle

repositioning and customer wait is to ask customers to walk a short distance to meet a stall at some specified location (which we refer to as a “rendezvous”), but this approach incurs customers’ hassle of traveling.

- A downside of the asset lightness is that stall vehicles generally have limited capacity and cannot carry too much inventory. In the presence of high demand, inventory replenishment may have to occur frequently and offsets the cost advantage of asset lightness.

Through our study, based on the merits, the challenges, and the ongoing pilot implementations, we aim to understand whether and how to practice the self-driving-enabled stall economy from an operations research perspective. We considered a setting in which a retailer deploys and operates a fleet of unmanned wheeled stalls for selling products (e.g., groceries) over an urban region. The stalls can be considered either as an independent retail network or as a complement to a brick-and-mortar store (which also serves as a depot). On the basis of such a setting, this paper presents models, theory, and insights on spatial queueing systems in which mobile stalls move around to meet mobile customers. From a practitioner’s perspective, we aim to understand how the retailer can strike a balance between vehicle repositioning costs, customers’ disutility of waiting and walking, and inventory replenishment expenses.

These results are presented in the following sequence.

We first developed models to examine how customers are served. To examine how each stall serves customers within a service area, we developed spatial queueing models to study two operating modes: (1) the on-demand mode, which assumes that the stall that finishes serving a customer in the current location always heads to the next customer who has waited the longest in the queue of online reservations, and (2) the demand-pooling mode, which assumes that the vehicle stays in each subarea for at least a guaranteed service duration before moving to the next subarea following a fixed rotation sequence. In addition, for inventory replenishment, we contrasted the conventional mode, where trucks visit a fixed set of locations, with the stall economy’s efficient inventory routing decisions that allow for jointly routing mobile stalls toward trucks to minimize the total replenishment cost—a salient feature of the stall economy. We developed a continuous approximation (CA) model to analytically characterize the optimal scheme of such joint routing. Integrating these models engendered an optimization model that can be used to prescribe deployment and operations of mobile stalls. We also considered managerial implications of the stall economy, including the value of stall mobility and its economic scalability.

The main contributions of this paper are as follows.

1. **Relevance.** This research is motivated by a high-level quest. How will the rapidly advancing self-driving technologies fundamentally reshape the future of retail

and the future of urban logistics (beyond mere last-mile deliveries)? To our knowledge, so far, the vast literature on retail operations has been within the scope of brick-and-mortar stores and the internet, and the literature on urban retail logistics has focused on last-mile deliveries. Departing from the literature, we envision the rise of autonomous mobile stalls in urban open space as the third pole of future retail and that these stalls may emerge as a new logistics form. Because this business model has already been piloted by major retail players, such as Stop & Shop, it is imperative to study whether and how to scale up the stall economy operations. As an initial attempt to study this problem, this work may stimulate future studies to explore a rich set of research questions implied by this new retail and logistics context.

2. Theoretical contributions. Innovations in urban transportation and logistics urge more studies on spatial queueing systems in which entities are mobile. This paper proposes and analyzes two spatial queueing systems in which one server moves around to meet mobile customers/machines and in which the “last 100 meters” are expensive. (i) First, this paper proposes and solves a *rendezvous problem* for the on-demand service mode. The problem involves finding the stall’s repositioning trip length, whereby the customer to serve is asked to walk toward a rendezvous point to meet the stall so that the stall avoids the last leg of the journey. Unlike the door-to-door setup, in which the stall destination exactly coincides with the customer location, the rendezvous problem is challenging because of the need to derive the spatial distribution of stall locations inherent in the rendezvous problem. This spatial distribution varies based on the choice of the rendezvous point. In addition, even if the rule of choosing the rendezvous is fixed, the problem is still complex because both the service time and the waiting time depend on the current “state,” where the state is defined as the locations and orders of all customers in the system. We model the transition of rendezvous as a Markovian system and propose a rule for picking the rendezvous, which we call the “magnet rule.” We show that, surprisingly, the stall distribution is uniform within an inner area when the demand density is homogenous over the entire service zone. (ii) Second, this paper also proposes and analyzes a spatial queueing system that pools spatially and temporally adjacent customer demands. We characterize the dependence of the customer mean waiting time on key decisions, including the promised service time in each subzone.

3. Insights. A major finding of this paper is that other things being equal, the stall economy can be more profitable than stationary retail. This advantage mainly results from the stall’s mobility, which provides proximity to customers (as well as saving the inventory routing cost). The efficiency of the stall economy is also partly attributable to its operational flexibilities. In the

presence of low customer demand, stalls can function in the on-demand, within-radius (or even door-to-door) mode to swiftly reposition to the next customer. As the demand scales up, the on-demand mode should give way to the demand-pooling mode to avoid excessive repositioning and customers’ wait and walk.

4. The paper also sheds light on how customers’ perception or reception of service quality poses challenges to stall economy operations. If customers become less willing to wait or walk, the stall retailer has to forego some benefits of demand pooling to avoid lost sales and must switch to the more costly on-demand mode in some moderate-demand areas. In contrast, engaging customers at stalls for a longer time increases the advantage of demand pooling. These insights also highlight the value of flexible stall operations compared with stationary retail.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces the model setup and presents three delivery service modes. Section 4 presents the integrated stall deployment model. Section 5 shows case studies and managerial insights. Section 6 concludes the paper. Additional proofs and parameter calibrations are provided in the online appendices.

2. Literature

Very few studies have focused on the emerging problem of *stall economy operations*. The vast literature on retail operations thus far has fallen within the scope of brick-and-mortar stores and the internet. We refer readers to Gallino and Moreno (2019) and Caro et al. (2020) for a comprehensive review of the achievements and challenges in the research on retail operations. To our knowledge, Glaeser et al. (2019) is the only paper in the extant literature that considers the operations of mobile retail. In their setting, an online retailer sends a fleet of trucks from which customers pick up their online orders. They combine empirical estimation of cannibalization effects, a random forest algorithm for demand prediction, and an integer program to solve for the optimal truck locations and schedule. In contrast to their assumptions of using trucks for online order fulfillment and of decisions being made on a weekly or monthly basis, our paper considers mobile stalls as a retail channel to satisfy customer demands that pop up in real time. To effectively satisfy these demands, the retailer needs to evaluate stall repositioning strategies (to reach customers without much delay) and inventory replenishment strategies, both of which are incorporated as the unique features of our models.

Our work on dispatching vehicles as mobile stores is related to the growing literature on dispatching couriers for last-mile deliveries, including food delivery service (Mao et al. 2019, Liu et al. 2021, Chen et al. 2022, Feldman et al. 2022). The majority of the papers

on last-mile deliveries assume that each vehicle trip delivers multiple orders. For example, Reed et al. (2022) consider using autonomous vehicles to assist human couriers. They identify significant reductions in completion time by avoiding parking. Liu et al. (2021) integrate travel time predictors (which are derived from real data based on drivers' routing behavior) with order assignment optimization. Qi et al. (2018) and Cao et al. (2020) consider crowdsourcing in idle ride-share mobility. In these contexts, the main focus is on formulating, solving, and/or approximating a vehicle routing problem of visiting a set of locations that are prespecified before a courier sets out. Fewer papers consider on-demand deliveries involving one or two orders per trip. For example, Chen and Hu (2020) study a courier dispatching problem in an on-demand delivery system and evaluate the effect of temporally pooling two consecutive orders. In contrast to their setting, in which the courier always sets out from the center of the service area, we focus on stall movement strategies and on how the optimal operational strategy (either the on-demand mode or the demand-pooling mode) differs among regions. We also incorporate the operating cost, which critically hinges on the mobility of stalls.

The spatial queueing models developed in our paper are related to the expanding literature on ride-hailing systems, where a platform matches riders with drivers. Spatial queueing is a critical factor in designing the matching system (Banerjee et al. 2015, 2021; Bimpikis et al. 2019; Benjaafar et al. 2022b). In particular, Besbes et al. (2022) study a spatial multiserver system that investigates the relationship between capacity and performance for a service firm with spatial operations. Our work shares a similar spirit with these works in the sense that our models also integrate spatial incentive and capacity issues. However, our modeling is driven by features that are unique to the stall economy operations: the issues of "avoiding the last 100 meters" (which motivates the rendezvous problem) and pooling temporally and spatially adjacent demands. For other recent studies that analyze on-demand systems, we refer readers to Cachon et al. (2017), Castillo et al. (2017), Taylor (2018), Gurvich et al. (2019), Yildiz and Savelsbergh (2019), Feng et al. (2021), and Benjaafar et al. (2022a).

3. Stall Economy Service Operations

This section models how mobile stalls serve customers. We first introduce the model setup in Section 3.1 and then, describe three service modes in Sections 3.2–3.4. A summary of notation is provided in Online Appendix A. Technical proofs are available in Online Appendix B.

3.1. Setup

3.1.1. Spatial Configuration. Consider a retailer running a fleet of autonomous mobile stalls across an urban

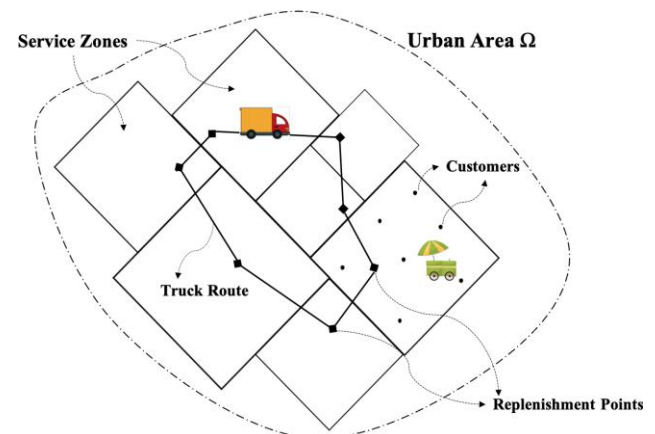
area Ω . The entire area is partitioned into rhombus-shaped service zones, as shown in Figure 2. Each stall sells products (e.g., groceries) within one service zone. Determining the spatial allocation of stalls is equivalent to tiling and sizing service zones. Assume that customer demands constitute a point process. The arrival rate around location $x \in \Omega$ and at hour $t \in \mathcal{T}$ (where $\mathcal{T}$ is the set of operating hours of a day) is $\lambda(x, t)$ (per hour per kilometer squared). To account for the accessibility of the mobile stalls, we assume that a stall is able to serve a customer only if it is in proximity to the customer. Otherwise, the customer has to wait until the stall moves to a nearby place. The moving distances (or interchangeably termed trip lengths) of stalls and customers are all in the l_1 norm (i.e., the Manhattan metric) throughout the paper.

This problem setting implies a fundamental trade-off. On the one hand, the larger the service zones are, the fewer stalls are needed to cover the entire urban area. On the other hand, larger service zones imply a longer wait for customers and more repositioning trips for stalls to reach customers. Customers' waiting time may affect their willingness to purchase at a mobile stall.

Moreover, stalls are capacitated. A truck tours around the area Ω to replenish the stalls' inventory. The retailer can exploit the mobility of the stalls to save the routing cost. Specifically, instead of moving to the exact location of a stall, the truck can move to a nearby location, whereas the stall moves toward the truck. After finishing the replenishment, the stall moves back to serve customers in its service zone. Such joint routing of the truck and stalls can potentially save the total fuel and wage costs.

3.1.2. Objective and Decisions. The retailer's objective is to maximize the profit from the stall economy operations. The profit equals the profit from selling products, minus the cost of purchasing stall vehicles (and permits to be on the street), and the operating costs of dispatching the stalls and the truck. The main decision is

Figure 2. (Color online) An Illustration of Citywide Mobile Retail Operations



the stall deployment (i.e., the layout of service zones over the urban area Ω). In the meantime, the retailer needs to make a repositioning scheme for each stall within a service zone to satisfy customer demands and to route the truck and stalls for inventory replenishment.

The stall deployment problem considered in this paper adopts the CA approach, which was first proposed by Newell (1971) and is designed to overcome the computational challenge associated with large-scale, discrete, combinatorial problems by approximating the objective with simpler mathematical formulas. That is, we do not intend to seek the exact delimitation of service zones by solving integer programming-based models; this computation generally is cumbersome, and insights are difficult to generate. Instead, we seek to optimize the density of stalls denoted by $\rho(x)$ (i.e., the number of stalls deployed per kilometer squared around any location x). This CA approach allows us to analytically capture the main trade-offs and to obtain concise solutions that lead to key managerial insights. CA has also been used in the context of logistics systems (Beardwood et al. 1959; Daganzo 2005; Carlsson et al. 2016, 2018), supply chains (Lim et al. 2017), and retail stores' layout (Cachon 2014, Belavina 2021). For an overview of the CA approach, we refer readers to Ansari et al. (2018). Our paper contributes to this line of research by combining elements from the CA paradigm together with Markovian queueing theory.

Under this setup, in the following sections we focus on a single service zone and examine three different modes of operating a stall to serve customers. The key operational metrics that we aim to characterize for each mode are the stall repositioning trip length and the customer waiting time. In the remainder of this section, the location index x or the time index t is suppressed from symbols wherever the discussion holds for all locations or at all times. The first two modes we consider are both *on demand*.

3.2. On Demand and Door to Door

In on-demand modes, each stall moves to reach customers in the order in which they request shopping via their smartphone (i.e., first come, first served (FCFS)). Such a spatial queueing setting can be modeled as an M/G/1 queue (in which demand arrivals are Markovian, service times follow a general distribution, and there exists a single server) if customers' requests occur following a Poisson process around location x and time t . Given the density of stalls $\rho(x)$ around x , each stall serves a zone of area $1/\rho(x)$. The demand arrival rate seen by the stall can thus be approximated by $\lambda(x)/\rho(x)$. Here, the mean demand density $\lambda(x)$ is typically a slowly varying function in x across the city and thus, can be assumed to be locally uniform around any location x .

The service time that the stall spends on each customer consists of both the time of repositioning from the previous location (i.e., where the stall served the

previous customer) and the time spent on the spot serving the customer. The repositioning time depends on the proximity at which the stall is required to reach customers. Specifically, we start with a *door-to-door* scenario in this section and discuss a *within-radius* scenario in the next section.

We use $L(\rho, r)$ to denote the repositioning trip length given stall density ρ and the consumer's walking distance r . In the door-to-door scenario, the stall moves to the doorstep of each customer, so that $r = 0$. Hence, the repositioning trip distance, $L(\rho, 0)$, is the distance between two random demand points in the service zone. Under the assumption that the mean demand density λ is uniform in the service zone, the following proposition provides the mean and the variance of this repositioning trip length in a service zone with edge length $R(\rho) = 1/\sqrt{\rho}$. We would like to emphasize that this zone edge length is naturally defined in the l_2 (Euclidean) norm (and thus, differs from the l_1 trip distance between two neighboring corners of a service zone, $\sqrt{2/\rho(x)}$). For ease of notation, we use L_r for fixed ρ in this section.

Proposition 1. *The mean and the variance of the door-to-door repositioning trip length are*

- i. $\mathbb{E}[L_0] = \frac{7\sqrt{2}R(\rho)}{15}$ and
- ii. $\text{Var}(L_0) = \frac{22R(\rho)^2}{225}$.

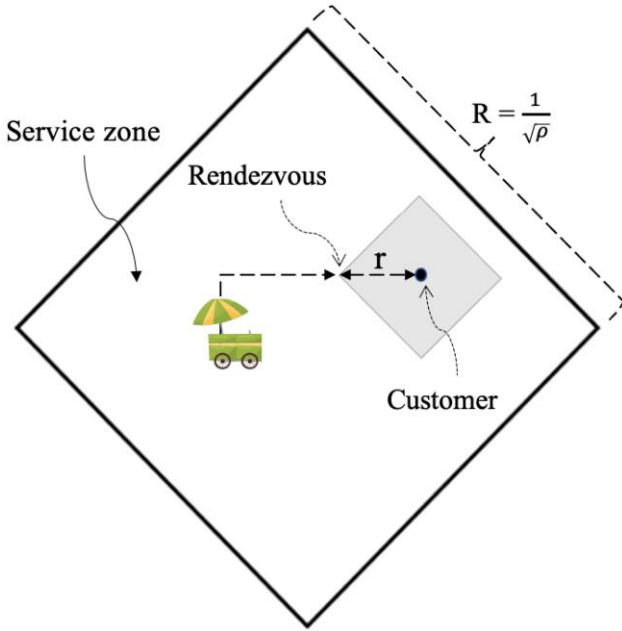
We discuss the waiting time implied by this service mode later in the next section.

3.3. On Demand and Avoiding the “Last 100 Meters” (the Rendezvous Problem)

In reality, regulations, roads, and building configurations may impede retail stalls from accessing customer doorsteps so that customers need to undertake the “last 100 meters” to meet a stall. Retailers also have an incentive to reduce or shorten repositioning trips and customer waiting times. These considerations motivate us to generalize the door-to-door scenario to a *within-radius* scenario. That is, we assume that customers are asked to take a walk of up to r distance to meet the stall. Thus, the stall either does not need to reposition if the next customer is already within the r radius, or it needs to move toward the customer until they are r distance apart, as Figure 3 illustrates.

We call the problem of finding the mean and the variance of the stall's repositioning trip length L_r in the within-radius scenario the *rendezvous problem*. Here, “rendezvous” refers to the location where the stall and a customer arrange to meet. Compared with the special case in which $r = 0$ in the door-to-door scenario, this rendezvous problem scenario generally is much more difficult to solve. Whereas the rendezvous locations for $r = 0$ coincide with customer locations and are thus uniformly and independently distributed in the service zone, if $r > 0$, the distribution of these locations is no

Figure 3. (Color online) An Illustration of the Within-Radius Service Mode



Note. Stall and customer trip distances are in the l_1 norm, and the service zone edge length R is in the l_2 norm.

longer obvious; in this case, they constitute a Markov chain. The rendezvous to serve the next customer depends on the rendezvous for the current customer. We need to identify the stationary distribution of rendezvous locations from their Markovian transitions before we are able to compute the mean and the variance of L_r .

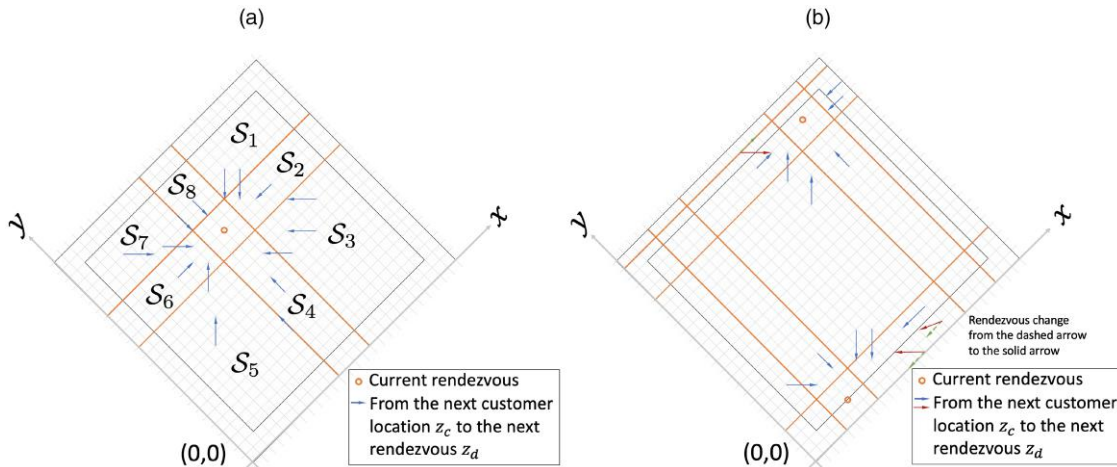
3.3.1. The Markov Chain Setup. We next tackle the challenge of finding the stationary distribution of

rendezvous locations. For the ease of representing the Markov chain, we create two axes along the lower edges of the service zone, as shown in Figure 4. We first discretize the whole space to $k \times k$ grids, where k is an arbitrarily large integer. (It can be rescaled to the actual length R without loss of generality.) By restricting the mobile stall to dock at grid points, we prove that the stationary distribution is uniform for any k . In the last step, we take k to infinity to reach the conclusion. Under this setup, the ranges of the x axis and the y axis are both $[0, k]$. Each rendezvous transition involves three relevant locations: (i) the stall's original location (the current rendezvous): $z_o = (x_o, y_o)$; (ii) the next customer's location: $z_c = (x_c, y_c)$; and (iii) the next rendezvous: $z_d = (x_d, y_d)$. The service zone is discretized so that z_o, z_c , and z_d appear only at grid points, with their x and y coordinates being integers from $\{0, \dots, k\}$. Note that the stall can serve all customers by moving only within an inner diamond area that is r -distance inward from the boundary of the service zone (hereafter referred to as the inner diamond or $\mathcal{I}_0$). Moving out of this inner diamond is unnecessary and would incur extra repositioning trips back to the inner diamond. In other words, $z_o, z_d \in \mathcal{I}_0 := [\varphi k, k - \varphi k] \times [\varphi k, k - \varphi k]$, where $\varphi = r/(\sqrt{2}R) \leq \frac{1}{2}$. Note that we pick walking distance $r \in \mathbb{R}^+$ such that $\varphi = r/(\sqrt{2}R)$ is a rational number. Any rational number can be represented as $\varphi = m_1/m_2$, where m_1 and m_2 are both integers. Thus, $k = l \cdot m_2$ with any integer l satisfies that both k and φk are integers.

3.3.2. The “Magnet Rule” of Choosing Rendezvous.

Observe that if the stall's origin z_o and the customer location z_c are not along the same grid line, then more than one candidate location can serve as the rendezvous z_d (with the Manhattan metric of distance). We define a rule of picking rendezvous as a mapping from

Figure 4. (Color online) The Magnet Rule in the Rendezvous Problem



Notes. The tails of solid arrows represent the next customer location z_c , and the heads of solid arrows represent the next rendezvous z_d . (a) Magnet rule for a type 1 origin. (b) Magnet rule for type 2 origins.

(z_o, z_c) to z_d . Clearly, different rules result in different rendezvous distributions. Instead of examining all possible rules, we focus on one rule in particular, which we refer to as the “magnet rule.” A formal definition of this rule is given; it consists of two cases depending on the type of the stall’s original location z_o .

3.3.2.1. Type 1 Origin. We call a stall location (x, y) a type 1 location if $(x, y) \in \mathcal{I}_1$, where $\mathcal{I}_1 = [2\phi k, k - 2\phi k] \times [2\phi k, k - 2\phi k]$. In other words, a type 1 location z_o is an inner location so that the small diamond centering at that location with radius r , denoted by $\mathcal{I}(z_o) = [x_o - \phi k, x_o + \phi k] \times [y_o - \phi k, y_o + \phi k]$, is contained by the inner diamond $\mathcal{I}_0$.

When the stall’s origin z_o is a type 1 location, we define the magnet rule using the following graph representations. (1) If we draw the small diamond $\mathcal{I}(z_o)$, Figure 4(a) shows that the four extended lines along this diamond’s four edges divide the whole service zone into eight regions (in addition to $\mathcal{I}(z_o)$ itself):

$$\begin{aligned} \mathcal{S}_1(z_o) &:= \{(x, y) : x > x_o + \phi k, y > y_o + \phi k\}, \\ \mathcal{S}_2(z_o) &:= \{(x, y) : x > x_o + \phi k, y_o - \phi k \leq y \leq y_o + \phi k\}, \\ \mathcal{S}_3(z_o) &:= \{(x, y) : x > x_o + \phi k, y < y_o - \phi k\}, \\ \mathcal{S}_4(z_o) &:= \{(x, y) : x_o - \phi k \leq x \leq x_o + \phi k, y < y_o - \phi k\}, \\ \mathcal{S}_5(z_o) &:= \{(x, y) : x < x_o - \phi k, y < y_o - \phi k\}, \\ \mathcal{S}_6(z_o) &:= \{(x, y) : x < x_o - \phi k, y_o - \phi k \leq y \leq y_o + \phi k\}, \\ \mathcal{S}_7(z_o) &:= \{(x, y) : x < x_o - \phi k, y > y_o + \phi k\}, \text{ and} \\ \mathcal{S}_8(z_o) &:= \{(x, y) : x_o - \phi k \leq x \leq x_o + \phi k, y > y_o + \phi k\}. \end{aligned}$$

(2) To exhibit the magnet rule with respect to a pair of stall and customer locations (z_o, z_c) , we use an arrow $\langle z_c, z_d \rangle$ that points from z_c to the chosen rendezvous z_d , representing the customer movement. For ease of notation, we define $a^i(z)$ to be an arrow that starts from z and points in one of eight directions with length r (l_1 norm). Specifically,

$$\begin{aligned} a^1(z) &= \langle (x, y), (x - \phi k, y - \phi k) \rangle, \\ a^2(z) &= \langle (x, y), (x - \phi k, y) \rangle, \\ a^3(z) &= \langle (x, y), (x - \phi k, y + \phi k) \rangle, \\ a^4(z) &= \langle (x, y), (x, y + \phi k) \rangle, \\ a^5(z) &= \langle (x, y), (x + \phi k, y + \phi k) \rangle, \\ a^6(z) &= \langle (x, y), (x + \phi k, y) \rangle, \\ a^7(z) &= \langle (x, y), (x + \phi k, y - \phi k) \rangle, \text{ and} \\ a^8(z) &= \langle (x, y), (x, y - \phi k) \rangle. \end{aligned}$$

We use $a^i(z)[1]$ and $a^i(z)[2]$ to denote the tail and head of arrow $a^i(z)$, respectively. Now, we can define the first part of the magnet rule as follows.

Definition 1 (Magnet Rule for Type 1 Origins). Given a type 1 stall origin z_o and any customer location z_c , the rendezvous z_d is uniquely specified by

- I. $z_d = a^i(z_c)[2]$, which is the head of arrow $a^i(z_c)$, if $z_c \in \mathcal{S}_i(z_o), i = 1, \dots, 8$ and
- II. $z_d = z_o$ if $z_c \in \mathcal{I}(z_o)$.

Intuitively, the magnet rule for a type 1 stall origin picks the rendezvous so that the customer (if she is

outside the small diamond $\mathcal{I}(z_o)$) moves in one of the eight directions (west, northwest, north, etc.) so as to minimize her distance to the stall’s original location z_o . For example, for a customer location $z_c \in \mathcal{S}_1(z_o)$, the arrow $a^1(z_c) = \langle z_c, (x_c - \phi k, y_c - \phi k) \rangle$ points to the chosen rendezvous z_d . The stall does not move if the next customer is within radius r . Figure 4(a) depicts more examples where the tail of a solid arrow represents the next possible customer location z_c and the head of a solid arrow represents the next rendezvous z_d . Notice that if z_c is on one of the four extended lines (orange solid lines), the arrow’s direction is also along that line.

We use $\mathcal{O}(z_o)$ to denote the set of all arrows that conform to the magnet rule for each stall’s original location z_o . For a type 1 stall origin $z_o \in \mathcal{I}_1$, this set is given as follows:

$$\mathcal{O}(z_o) = \{a^i(z_c) \text{ if } z_c \in \mathcal{S}_i(z_o), i = 1, \dots, 8\}.$$

3.3.2.2. Type 2 Origin. Alternatively, we call a stall location (x, y) a type 2 location if $(x, y) \in \mathcal{I}_0 \setminus \mathcal{I}_1$. In other words, type 2 locations are “border” locations z_o such that their associated small diamonds $\mathcal{I}(z_o)$ (i.e., centering at z_o and with radius r) exceed the boundary of the inner diamond $\mathcal{I}_0$, as Figure 4(b) illustrates. The magnet rule for a type 2 stall origin z_o is a modification of the rule in Definition 1.

Definition 2 (Magnet Rule for Type 2 Origins). Given a type 2 stall origin $z_o = (x_o, y_o)$ and any customer location $z_c = (x_c, y_c)$, the rendezvous z_d is uniquely specified as follows.

I. If $z_c \in \mathcal{S}_i(z_o), i = 1, \dots, 8$, let $a^i(z_c)$ also be denoted by $\langle z_c, \hat{z}_d \rangle$, where the arrowhead $\hat{z}_d$ has coordinates $(\hat{x}_d, \hat{y}_d)$. Then, one of the following three cases applies:

- a. $z_d = \hat{z}_d$ if $\hat{z}_d \in \mathcal{I}_0$;
- b. $z_d = (\hat{x}_d, y_o)$ if $\hat{y}_d < \phi k$ or $\hat{y}_d > k - \phi k$; or
- c. $z_d = (x_o, \hat{y}_d)$ if $\hat{x}_d < \phi k$ or $\hat{x}_d > k - \phi k$.

II. If $z_c \in \mathcal{I}(z_o), z_d = z_o$.

Notice that the conditions in (Ib) and (Ic) cannot hold simultaneously. For example, if $\hat{x}_d < \phi k$ and $\hat{y}_d < \phi k$, then (Ib) and (Ic) together would imply that $z_d = (\hat{x}_d, y_o) = (x_o, \hat{y}_d)$ or $(x_o, y_o) = (\hat{x}_d, \hat{y}_d)$, which is impossible because z_o is in $\mathcal{I}_0$.

Definition 2 specifies that, if application of the rule in Definition 1 results in an arrow that points to a location within the inner diamond $\mathcal{I}_0$, then the proposed rendezvous remains valid. However, if a customer location z_c outside the inner diamond $\mathcal{I}_0$ points to a rendezvous that is also outside the inner diamond, then the rendezvous is changed to avoid moving the stall out of the inner diamond while ensuring that z_d is still r apart from z_c . Figure 4(b) illustrates two examples, in which we rotate the green dashed arrow to the red solid arrow for this purpose. Following this rule, we also can construct a set of all valid arrows, $\mathcal{O}(z_o)$, for a type 2 stall origin.

The magnet rule, despite its heuristic nature, boasts two important merits.

i. Optimality. The stall's repositioning trip distance to meet the next customer is guaranteed to be minimum, as formalized in the following proposition.

Proposition 2. *Under the Manhattan metric of distance and under the on-demand service mode, the magnet rule ensures that the stall travels the shortest distance to meet the next customer.*

We want to emphasize that there are other natural policies that achieve the shortest travel distance for one trip in the l_1 norm, but other policies may perform differently over multiple trips. In the numerical studies in Section 5.1, we will show that the magnet rule outperforms some other greedy rule of selecting rendezvous but is suboptimal relative to the clairvoyant scenario, in which future customer locations were known. However, these performance gaps are moderate.

ii. Tractability. Moreover, the magnet rule allows us to tractably solve the rendezvous problem, which we show as follows.

3.3.3. Transition and Stationary Probabilities. Based on the magnet rule of choosing the rendezvous, we next investigate the transition probability $\mathbb{P}(z_o, z_d)$ —that is, the probability that the next rendezvous is z_d given that the current rendezvous is z_o , $\forall z_o, z_d \in \mathcal{I}_0$. Define $\mathcal{A}(z_o, z_d)$ as the set of customer movement arrows that point at z_d and that belong to $\mathcal{O}(z_o)$. Because the service zone has a total of $(k+1)^2$ grid points as customer locations, the cardinality of $\mathcal{A}(z_o, z_d)$ satisfies $|\mathcal{A}(z_o, z_d)| = (k+1)^2$ for any $z_o \in \mathcal{I}_0$. In addition, because customers are assumed to be uniformly distributed among these grid points, we obtain $\mathbb{P}(z_o, z_d) = |\mathcal{A}(z_o, z_d)| / (k+1)^2$. With this construction of the transition probability, we obtain the following result.

Theorem 1. *The magnet rule leads to the following properties.*

I. *The transition matrix of the rendezvous's Markov chain is doubly stochastic—that is, both its row sum $\sum_{z_d \in \mathcal{I}_0} \mathbb{P}(z_o, z_d)$ and its column sum $\sum_{z_o \in \mathcal{I}_0} \mathbb{P}(z_o, z_d)$ are equal to one, $\forall z_o, z_d \in \mathcal{I}_0$.*

II. *The stationary distribution of the rendezvous is uniform over the inner diamond $\mathcal{I}_0$.*

Theorem 1 is one of the major technical results of this paper. The result is quite surprising because the uniformity is preserved over the inner area, although the stall locations are no longer uniformly distributed over the entire service zone. To be more specific, it shows that, following a properly defined rule of selecting rendezvous, the simple form of the rendezvous distribution (i.e., the uniform distribution) is still preserved when $r > 0$, despite two complexities. The support of this distribution shrinks from the entire service zone to the inner diamond $\mathcal{I}_0$; the rendezvous become temporally dependent. Underlying our proof of the unity of

the transition matrix's column sum (which is the key part of Theorem 1) is the following process. Given a rendezvous z_d in $\mathcal{I}_0$, we examine each location $z_o \in \mathcal{I}_0$ as the stall's origin. According to the magnet rule, a customer location z_c lets the stall move from z_o to z_d if and only if the customer movement arrow that starts from z_c and points at z_d belongs to $\mathcal{O}(z_o)$. By identifying these arrows, we construct $\mathcal{A}(z_o, z_d)$. We also can verify that the column sum $\sum_{z_o \in \mathcal{I}_0} \mathbb{P}(z_o, z_d)$ equals one by examining the cardinality of $\mathcal{A}(z_o, z_d)$. The arrow identification process varies depending on whether z_d is a type 1 or type 2 location and depending on the location of z_o relative to z_d . We provide the detailed proof in Online Appendix B.

Theorem 1 holds for any integer k satisfying the condition that φk is an integer ($\varphi = r/(\sqrt{2}R)$). Therefore, by incrementing k to infinity, we recover the continuous setting and conclude that the rendezvous are uniformly distributed inside $\mathcal{I}_0$. In the numerical studies in Section 5.1 and Online Appendix C, we will further show that this tractable uniform distribution is not necessarily achieved by other reasonable policies.

3.3.4. Repositioning Trip Distance. Having shown that the rendezvous are uniformly distributed in the inner diamond $\mathcal{I}_0$, we are now able to characterize the repositioning trip distance L_r .

Proposition 3. *To be within radius r ($0 \leq r \leq R/\sqrt{2}$) from a customer, the repositioning trip distance must satisfy the following:*

$$\text{I. } \mathbb{E}[L_r] = \frac{7\sqrt{2}R^3 - 22R^2r + 2\sqrt{2}Rr^2 + 12r^3}{15R^2},$$

$$\text{II. } \text{Var}(L_r) = \frac{22R^6 - 22\sqrt{2}R^5r - 140\sqrt{2}R^3r^3 + 400R^2r^4 - 48\sqrt{2}Rr^5 - 144r^6}{225R^4}, \text{ and}$$

III. *$\mathbb{E}[L_r]$ and $\text{Var}(L_r)$ are both increasing in R and decreasing in r .*

Proposition 3 completes the solution of the rendezvous problem by providing closed-form expressions for $\mathbb{E}[L_r]$ and $\text{Var}(L_r)$. Proposition 3 quantifies the first effect of avoiding the last “100 meters.” That is, increasing r alleviates the burden of the repositioning operations. These expressions also can be conveniently used for further modeling and analysis, as the remainder of this paper demonstrates.

3.3.5. Customer Waiting Time. We can now model the mean customer waiting time based on Proposition 3. We denote the stall's moving speed by v . The repositioning time, then, is L_r/v . Further, suppose that v is a random variable and that the first two moments of its inverse $\mathbb{E}[\frac{1}{v}]$ and $\mathbb{E}[\frac{1}{v^2}]$ are known. This randomness can account for the uncertainties of local traffic conditions, and we ignore any dependence of this randomness on the trip length L_r . After the stall arrives, suppose that the time spent on the spot serving a customer is another independent random variable with

mean $1/\mu$ and variance σ^2 . Then, the total service time for one customer, denoted by T , has a mean and variance as follows:

$$\mathbb{E}[T] = \mathbb{E}[L_r] \mathbb{E}\left[\frac{1}{v}\right] + \frac{1}{\mu}, \quad (1a)$$

$$\text{Var}(T) = \mathbb{E}[L_r^2] \mathbb{E}\left[\frac{1}{v^2}\right] - \mathbb{E}[L_r]^2 \mathbb{E}\left[\frac{1}{v}\right]^2 + \sigma^2, \quad (1b)$$

where $\mathbb{E}[L_r^2] = \mathbb{E}[L_r]^2 + \text{Var}(L_r)$. Hence, the expected waiting time of each customer is given by

$$\begin{aligned} w &= \frac{\frac{\lambda}{\rho} \mathbb{E}[T] + \frac{\lambda \text{Var}(T)}{\rho \mathbb{E}[T]}}{2 \left(\frac{1}{\mathbb{E}[T]} - \frac{\lambda}{\rho} \right)} + \mathbb{E}[L_r] \mathbb{E}\left[\frac{1}{v}\right] \\ &= \frac{\lambda (\text{Var}(T) + \mathbb{E}[T]^2)}{2(\rho - \lambda \mathbb{E}[T])} + \mathbb{E}[L_r] \mathbb{E}\left[\frac{1}{v}\right], \end{aligned} \quad (2)$$

in which the customer demand arrival rate is λ/ρ . The first term on the right-hand side of (2) is the in-line waiting time due to the Pollaczek–Khinchine formula and Little’s law (Shurtle et al. 2018, pp. 10 and 256). The second term is the expected repositioning time. Combining Proposition 3, Equations (1) and (2), and $R = 1/\sqrt{\rho}$ immediately reveals that the mean waiting time w is decreasing both in the stall deployment density ρ and in the radius r . Hence, the second effect of avoiding the last “100 meters” is a reduced customer waiting time, which enhances customer utility. We examine the endogeneity of the demand rate λ on customer utility in Section 4.

3.4. Demand Pooling

Although the on-demand mode is meant to provide express shopping service to customers, it may become neither “express” nor cost efficient if excessive repositioning must occur in the presence of high-volume and distributed demand. Motivated by this dilemma, we propose an alternate service mode called *demand pooling*, which pools demands that are adjacent both in time and in location to improve efficiency.

Specifically, we design a schedule for a mobile stall to visit N subzones (which span a service zone) sequentially and repeatedly to serve customers. The radius of each subzone is $r = R/\sqrt{2N}$. Suppose that customer demand arrivals in subzone i constitute a Poisson process with rate λ_i (so that $\lambda(x) \approx \lambda_i/(\text{area of subzone } i)$ if x is in subzone i).

The stall may be in another subzone at the time a customer places a request. Then, the customer waits until the stall arrives.

Once the stall arrives in a subzone, all waiting customers in that subzone start being served. Here, we assume that the stall can accommodate customers to allow for picking products in parallel, like at the grocery store. The stall will stay for a promised time t_i^p in subzone i . Here, t_i^p is a decision variable and is jointly optimized along with

the decisions of stall density and service radius in the stall economy planning problem in the next section. Suppose that the service time for each customer follows a distribution $G(\cdot)$ and is bounded within $[0, 1/\underline{\mu}]$. The promised stay must satisfy $t_i^p \geq 1/\underline{\mu}$ to ensure that at least the customers who were waiting can finish shopping. Moreover, all new customers in subzone i that arrive during $[0, t_i^p]$ also will be served. The process of serving these new customers forms an $M/G/\infty$ queue (in which the server capacity is ample). As shown in Figure 5, the actual stay in subzone i is divided into two stages: $[0, t_i^p]$ and $[t_i^p, \tau_i]$; the latter is the extended stay for finishing service to all customers who have arrived before t_i^p . Note that τ_i is random because of the uncertainty of arrival times and of service duration.

3.4.1. Customer Waiting Time. To understand the customer waiting time, we calculate $\mathbb{E}[\tau_i(t_i^p)]$ and $\mathbb{E}[\tau_i(t_i^p)^2]$ (i.e., the first two moments of the actual duration of stay τ_i as functions of the decision t_i^p).

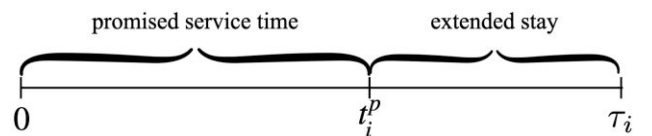
Let $Y_i(t)$ denote the number of customers being served in subzone i at time $t \geq t_i^p$. By time $t(\geq t_i^p)$, all customers who arrived before time 0 have already finished shopping. A customer who arrived at time $s \in [0, t)$ either finished the service with probability $G(t-s)$ or still remains in the system with probability $\bar{G}(t-s)$, where $\bar{G}$ represents the complementary cumulative distribution function ($\bar{G}(\cdot) = 1 - G(\cdot)$). Because of the Poisson split property, $Y_i(t)$ follows a Poisson distribution with mean $\mathbb{E}[Y_i(t)] = \lambda_i \int_0^{t_i^p} \bar{G}(t-s) ds$, which implies that $\mathbb{P}(\tau_i(t_i^p) \leq t) = \mathbb{P}(Y_i(t) = 0) = \exp\left(-\lambda_i \int_0^{t_i^p} \bar{G}(t-s) ds\right)$. Hence, the expected value of $\tau_i(t_i^p)$ becomes

$$\begin{aligned} \mathbb{E}[\tau_i(t_i^p)] &= \int_0^\infty \mathbb{P}(\tau_i(t_i^p) > t) dt \\ &= t_i^p + \int_{t_i^p}^{t_i^p + 1/\underline{\mu}} \left(1 - \exp\left(-\lambda_i \int_0^{t_i^p} \bar{G}(t-s) ds\right)\right) dt. \end{aligned} \quad (3)$$

Similarly, the second moment of $\tau_i(t_i^p)$ is given by

$$\begin{aligned} \mathbb{E}[\tau_i(t_i^p)^2] &= \int_0^\infty \mathbb{P}(\tau_i(t_i^p)^2 > t) dt = (t_i^p)^2 \\ &\quad + \int_{(t_i^p)^2}^{(t_i^p + 1/\underline{\mu})^2} \left(1 - \exp\left(-\lambda_i \int_0^{t_i^p} \bar{G}(\sqrt{t}-s) ds\right)\right) dt. \end{aligned} \quad (4)$$

Figure 5. Composition of the Service Time in Subzone i



Observe that this demand-pooling process is a renewal process, where the process ends and restarts each time the mobile stall revisits any subzone i . Let $T^c = T^r + \sum_{i=1}^N \tau_i(t_i^p)$ represent the cycle time, and let $T^r := \sum_{i=1}^N T_i^r$ denote the total repositioning time in a complete cycle, in which each T_i^r is the time it takes to reposition from subzone i to $i + 1$. Applying the renewal reward theorem yields the expected waiting time over all customers.

Theorem 2. *Given a promised duration of stay t_i^p for each subzone $i \in \{1, 2, \dots, N\}$, the average waiting time of customers in subzone i equals*

$$w_i(t_i^p) = \frac{h_i(t_i^p)/2}{\mathbb{E}[T^r] + \sum_{j=1}^N \mathbb{E}[\tau_j(t_j^p)]}, \quad (5)$$

where $h_i(t_i^p) := \mathbb{E}[(T^c - t_i^p)^2] = \mathbb{E}[(T^r)^2] + \sum_{j \neq i} \mathbb{E}[\tau_j(t_j^p)] \mathbb{E}[\tau_i(t_i^p)] + \sum_{j=1}^N \mathbb{E}[\tau_j(t_j^p)^2] + 2(\mathbb{E}[T^r] - t_i^p) \cdot \left(\sum_{j=1}^N \mathbb{E}[\tau_j(t_j^p)] - t_i^p \right) - (t_i^p)^2$ with $\mathbb{E}[\tau_i(t_i^p)]$ and $\mathbb{E}[\tau_i(t_i^p)^2]$ given by Equations (3) and (4).

Theorem 2 demonstrates the value of pooling. The waiting time w_i is less sensitive to the arrival rate λ_i in the demand pooling mode than in the on-demand mode. This can be understood by observing that the way that λ_i increases w_i is via $\mathbb{E}[\tau_i(t_i^p)]$ and $\mathbb{E}[\tau_i(t_i^p)^2]$. However, this impact diminishes rapidly because λ_i only appears in the exponentially decaying functions in Equations (3) and (4). To understand (5), observe that $\lambda_i h_i(t_i^p)/2$ represents the total waiting time of consumers in subzone i during a renewal cycle when the promised stay duration is t_i^p . Here, $h_i(t_i^p)$ in each subzone depends on the total repositioning time T^r and the time of stay in all subzones, and it can be expressed as $h_i(t_i^p) = \mathbb{E}[(T^c - \tau_i(t_i^p))^2]$. Meanwhile, $\lambda_i \mathbb{E}[T^c]$ is the expected number of customers who arrive during a cycle. Therefore, the arrival rate λ_i cancels out in the numerator and the denominator of (5), and it only shows up in the expected service time $\mathbb{E}[\tau_i(t_i^p)]$ and $\mathbb{E}[\tau_i(t_i^p)^2]$. Compared with the on-demand mode (Equation (2)), the waiting time here is more robust to the demand arrival rate in the demand pooling mode, which demonstrates the value of pooling.

Unfortunately, Theorem 2 does not admit a closed-form expression for the mean waiting time because the expressions for the first and second moments of $\tau_i(t_i^p)$ are determined by the service time distribution. Hence, we derive the following result for special cases where the service time is uniformly distributed or deterministic, and the demand arrival rates are homogenous across subzones.

Corollary 1. *The following two conclusions hold.*

I. *If the arrival rates are homogenous (i.e., $\lambda_i = \lambda_j = \lambda_c$ for any $i, j \in \{1, 2, \dots, N\}$) and the service time is uniformly*

distributed on $[0, 1/\mu]$, then the mean waiting time equals

$$w(t^p) = \frac{h(t^p)/2}{\mathbb{E}[T^r] + N\mathbb{E}[\tau(t^p)]}, \quad (6)$$

in which

$$\begin{cases} h(t^p) = \mathbb{E}[(T^r)^2] + N(N-1)\mathbb{E}[\tau(t^p)]^2 + N\mathbb{E}[\tau(t^p)^2] \\ \quad + 2(\mathbb{E}[T^r] - t^p) \cdot (N\mathbb{E}[\tau(t^p)] - t^p) - (t^p)^2, \\ \mathbb{E}[\tau(t^p)] = t^p + 1/\mu - \frac{\sqrt{\pi}}{2} \exp(c_1)(\operatorname{erfi}(c_2) - \operatorname{erfi}(c_3)), \\ \mathbb{E}[\tau(t^p)^2] = (t^p + 1/\mu)^2 - \exp(c_4) \left(\exp(c_6^2) - \exp(c_5^2) \right) \\ \quad + \frac{c_7 \sqrt{\pi}}{2} (\operatorname{erfi}(c_6) - \operatorname{erfi}(c_5)), \end{cases}$$

where $\operatorname{erfi}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{t^2} dt$ and constants $c_1 = -\lambda_c \left(\frac{\mu(t^p)^2}{2} + \frac{1}{2\mu} - \frac{1}{4} \right)$, $c_2 = \sqrt{\lambda_c}(t^p + 1/\mu + 1/2)$, $c_3 = \sqrt{\lambda_c}(t^p + 1/2)$, $c_4 = -\lambda_c \left(\frac{1}{2\mu} + \frac{\mu(t^p)^2}{2} \right)$, $c_5 = \sqrt{\lambda_c \left(\frac{\mu+2}{2} \right)} \left(t^p - \frac{1}{\mu+2} \right)$, $c_6 = \sqrt{\lambda_c \left(\frac{\mu+2}{2} \right)} \left(t^p + \frac{2}{\mu(\mu+2)} \right)$, and $c_7 = \sqrt{\frac{2\lambda_c}{\mu+2}}$.

II. *Furthermore, if the service time is deterministic and equal to $1/\mu$, then*

$$\begin{cases} \mathbb{E}[\tau(t^p)] = t^p + 1/\mu - \frac{1}{\lambda_c} (1 - \exp(-\lambda_c/\mu)), \\ \mathbb{E}[\tau(t^p)^2] = (t^p + 1/\mu)^2 - \frac{2}{\lambda_c} ((t^p + 1/\mu) \\ \quad - t^p \exp(-\lambda_c/\mu)) + \frac{2}{\lambda_c^2} (1 - \exp(-\lambda_c/\mu)). \end{cases} \quad (7)$$

Corollary 1 provides closed-form expressions for the expected waiting time: All terms on the right-hand side of (7) can be readily computed (notice that $N = 1/(2\rho r^2)$, $\mathbb{E}[T^r] = 2rN\mathbb{E}[\frac{1}{v}]$, and $\mathbb{E}[(T^r)^2] = 4r^2N^2\mathbb{E}[\frac{1}{v^2}]$).

3.4.2. Repositioning Trip Distance. Given N subzones in a service region, the retailer's goal is to maximize the total profit, which is the gross profit from sales net of the stall's cost over the operational time $\mathcal{T}$ during the day. One part of the cost is the stall's repositioning cost. Under the renewal reward theorem, the repositioning trip distance in the service zone per unit of time is

$$\frac{2rN}{\mathbb{E}[T^r] + \sum_{i=1}^N \mathbb{E}[\tau_i(t_i^p)]}, \quad (8)$$

where $2rN$ is the fixed trip length of traversing N subzones. Integrating (8) over one day yields the total repositioning trip distance:

$$l_s := \int_{t \in \mathcal{T}} \frac{2rN}{\mathbb{E}[T^r(t)] + \sum_{i=1}^N \mathbb{E}[\tau_i(t_i^p(t))]} dt, \quad (9)$$

where $T^r(\cdot)$ and $t_i^p(\cdot)$ denote the time-varying repositioning cost and the promised stay time, respectively.

3.4.3. Trade-Offs Between Waiting Time and Repositioning Trip Distance. The models developed in this section imply that the promised stay time t_i^p has conflicting effects on the repositioning trip distance and customer waiting time (1) Expression (8) shows that the repositioning trip distance per unit time decreases when t_i^p increases because τ_i is stochastically increasing in t_i^p . (2) Meanwhile, Theorem 2 implies that staying longer in subzone i also decreases the mean waiting time w_i for customers in i . The reasons are that increasing $\tau_i(t_i^p)$ increases the cycle time T^c , whereas $h_i(t_i^p)$ remains the same (because $h_i(t_i^p) = \mathbb{E}[(T^c - t_i^p)^2] = \mathbb{E}[(T^r + \sum_{j \neq i} \tau_j(t_j^p) + \tau_i(t_i^p) - t_i^p)^2]$ and the distribution of $\tau_i(t_i^p) - t_i^p$ is independent of t_i^p). It directly implies that the mean waiting time in subzone i decreases. (3) However, staying too long in a single subzone i may increase the waiting time of customers in other subzones. From the customers' perspective, the longer their waiting time is, the less willing they are to purchase. These conflicting effects motivate us to jointly optimize t_i^p along with stall deployment decisions. We address this optimization in Section 4.

To conclude our discussion of the rendezvous problem and the demand-pooling model, we remark that these technical results may represent a meaningful contribution because the setup of one server moving around to meet mobile customers/machines is increasingly prevalent in many on-demand services and in unmanned warehouse operations. We will briefly discuss possible variants of these spatial queueing systems toward the end of this paper.

4. The Stall Economy Planning Problem

Having studied the service operations, in this section we present the stall economy planning problem. Solving this problem needs to take into account not only the service operations described in the preceding section but also, the inventory routing cost (Section 4.1), the endogeneity of customer demands (Section 4.2), and the vehicle cost. Striking a balance among these factors yields the optimal stall deployment density at each location and the associated optimal profit (Section 4.3).

4.1. Routing for Inventory Replenishment

We model the operations of inventory replenishment. Because stalls have a limited capacity, assume that a truck is dispatched periodically to replenish the inventory of the stalls. We focus on exploiting the mobility of stalls to minimize the total traveling cost incurred by the truck and stalls. That is, the truck can take a shorter route than in the classical vehicle routing setting because stalls can travel the “last mile” to the truck for replenishment.

Let l denote the decision of the truck route selected from the set of all possible routes $\mathcal{L}$. Let $d(x, l)$ denote the shortest distance from a location $x \in \Omega$ to the route l . Suppose that a stall incurs c_s U.S. dollars (USD) per kilometer of travel and that the truck incurs c_t USD per kilometer of travel. Parameters c_s and c_t may account for vehicle operating costs and/or time-based costs (e.g., truck driver wages and extended customer wait). Given any stall deployment density $\rho(\cdot)$, the minimum joint routing cost $\text{RC}(\rho)$ for each round of replenishment is given by the following formulation:

$$\text{RC}(\rho) = \underset{l \in \mathcal{L}}{\text{minimize}} \quad c_t \|l\| + 2c_s \iint_{x \in \Omega} \rho(x) d(x, l) dx, \quad (10)$$

where the first term is the truck routing cost, $\|l\|$ denotes the length of route l , and the second term represents the stall routing cost (including both the forwarding and returning trips). We then obtain the following asymptotic result for the optimal routing cost.

Proposition 4. Suppose that the random locations of n stalls are independent and identically distributed on a compact planar region Ω according to an absolutely continuous probability density function $\rho(x)/n$. Then, with probability of one, the minimum joint routing cost $\text{RC}(\rho)$ satisfies

$$\lim_{n \rightarrow \infty} \frac{\text{RC}(\rho)}{\iint_{x \in \Omega} \sqrt{\rho(x)} dx} = \sqrt{2c_t c_s}. \quad (11)$$

The proof of Proposition 4 is based on a result in Carlsson and Song (2018) (see Online Appendix B). Given Proposition 4, we use

$$\sqrt{2c_t c_s} \iint_{x \in \Omega} \sqrt{\rho(x)} dx \quad (12)$$

to approximate the joint routing cost $\text{RC}(\rho)$.

4.1.1. The Value of Stall Mobility in Inventory Routing. To interpret Equation (12), consider a baseline in which the stalls cannot move. A truck route traversing the centers of all service zones would cost $\sqrt{2}c_t \iint_{x \in \Omega} \sqrt{\rho(x)} dx$ under the Manhattan metric. The reason is that the trip length of traversing one service zone centering at x is $\sqrt{2/\rho(x)}$, and there are locally $\rho(x)$ such zones per kilometer squared. Hence, letting the stalls move changes the routing cost to be $\sqrt{c_s/c_t}$ of the baseline level, and this change is cost effective if and only if $c_s < c_t$, which we assume in the remainder of the paper. (Otherwise, we could simply use the baseline routing cost.)

Note that Proposition 4 relies on several simplifications based on the following three rationales. (i) First, Proposition 4 considers a single truck. However, even under the generalized scenario, in which a finite number of trucks divides the area Ω and each truck meets a

subset of the stalls, the expression of the optimal joint routing cost in Proposition 4 still asymptotically holds. The proof is based on constructing modified service areas and vehicle routes and is omitted for brevity. (ii) Second, we assume the truck replenishes the stalls to their capacity. Our context typically involves small-sized stalls, fast inventory turnover, and a relatively sufficient supply of products. Therefore, not filling the stalls up to their capacity is not likely to be optimal. (iii) Third, although the optimal joint routing cost Equation (12) is exact only when $n \rightarrow \infty$, we use this formula because it approximately and conveniently captures the routing cost trade-offs. Similar results in a “horsefly” problem setting have been numerically verified by Carlsson and Song (2018) to hold for realistic problem instances with finite n .

4.1.2. Daily Routing Cost. Next, we estimate the daily joint routing costs for inventory replenishment. The previous discussions qualifying Proposition 4 suggest that a common replenishment frequency for all the stalls may not exist because multiple trucks may visit the stalls on different schedules. In addition, the inventory depletion rate may vary by location. Therefore, we approximate the replenishment frequency locally. For a stall around any location x , the average number of replenishments per day, denoted by $m(x)$, can be chosen so that the average vacant space to fill upon replenishment, denoted by S , is a certain proportion (e.g., 80%) of a stall’s loading capacity. Recall that $\lambda(x, t)$ denotes the arrival rate around location $x \in \Omega$ (per hour per kilometer squared) at time t . Hence, one can estimate $m(x)$ as follows:

$$m(x) = \frac{\int_{t \in \mathcal{T}} \lambda(x, t) s dt}{\rho(x) S}, \quad (13)$$

where s is the average volume of products purchased by a customer. Multiplying (11) by (13) yields the following estimation of the daily joint routing cost for inventory replenishment:

$$\sqrt{2c_t c_s} \iint_{x \in \Omega} \frac{\int_{t \in \mathcal{T}} \lambda(x, t) s dt}{\sqrt{\rho(x) S}} dx. \quad (14)$$

Although the truck-stall joint routing problem connects the service zones, (14) is additive across all locations. As a result, the stall deployment problem that integrates (14) can still be conveniently solved point by point.

4.2. Endogenous Customer Demand

The customer demand arrival rate λ depends on the customer utility of using mobile stalls. A long wait or a long walk to meet a stall decreases a customer’s utility and dampens his or her willingness to purchase. Specifically, let the ex ante net utility of using a mobile stall be $U(w, r) = v(w, r) + \epsilon$, where $v(w, r)$ is the intrinsic valuation of

purchasing from the stall with average waiting time w and walking distance r and where ϵ is the unobserved part, which is assumed to follow a Gumbel distribution. The function $v(w, r)$ decreases in the mean waiting time w and the service radius r . In our numerical studies, we assume $v(w, r)$ to be linear in w and r ; that is,

$$v(w, r) = \alpha_0 - \alpha_1 w - \alpha_2 r, \quad (15)$$

where α_0 , α_1 , and α_2 are constant nonnegative coefficients. Further, suppose that each consumer weighs two alternatives: either to purchase or not to purchase at a mobile stall. By normalizing the utility of no purchase to zero and assuming that each customer maximizes his or her utility, the selection probability is given by

$$q(w, r) = \mathbb{P}(U(w, r) \geq 0) = \frac{e^{v(w, r)}}{1 + e^{v(w, r)}}.$$

The demand arrival rate at location x and time t can thus be expressed as follows:

$$\lambda(x, t; w, r) = \lambda_0(x, t) \cdot q(w, r), \quad \forall x \in \Omega, t \in \mathcal{T}, \quad (16)$$

where $\lambda_0(x, t)$ is the intrinsic arrival rate of all potential consumers at location x and at time t , which is independent of w and r .

Equation (16) models the third effect of avoiding the “last 100 meters” as an unfavorable reduction in demand (in addition to the favorable effects of reducing repositioning and waiting). We address these conflicting effects in the next section by jointly optimizing the service radius and the stall deployment density.

4.3. Stall Deployment

The planning of the stall economy system is represented by decisions regarding both the stall deployment density $\rho(x)$ and the service radius $r(x)$ around each location x . The objective is to maximize the expected daily profit density (in dollars kilometer⁻²) at x expressed as follows:

$$\begin{aligned} R(x; \rho(x), r(x)) := & \underbrace{p \Lambda(x; \rho(x), r(x))}_{\text{profit from sales}} - \underbrace{c_s I_s(x; \rho(x), r(x))}_{\text{stall repositioning cost}} \\ & - \underbrace{c_p \rho(x)}_{\text{fixed cost}} - \underbrace{\sqrt{2c_t c_s} \frac{\Lambda(x; \rho(x), r(x)) s}{\sqrt{\rho(x) S}}}_{\text{inventory routing cost}}. \end{aligned} \quad (17)$$

The first term represents the profit from sales, where p is the average profit earned from a shopper. $\Lambda(x; \rho, r) = \int_{t \in \mathcal{T}} \lambda(x, t; w(\rho, r), r) dt$ denotes the density of total demand over one day at location x given stall density ρ and radius r . Notice that here we assume $\rho(x)$ (as well as $r(x)$) to be fixed during time horizon $\mathcal{T}$. To extend to the scenario where the stall deployment can change during the day, we can solve the optimization problem separately for different time horizons. For example, we can set $\mathcal{T}$ as morning hours and then, as afternoon hours. Nevertheless, redispacting stalls during the day to adapt

to the demand nonstationarity incurs extra cost, which we leave for future research. The second term represents the stall repositioning cost for serving customers, where $l_s(x; \rho, r)$ is the total repositioning trip length per kilometer squared around location x . The third term is the density of the distance-independent cost of running the stall fleet per day. Here, c_p is the cost per stall (e.g., the cost of a permit for being on the street). The fourth term comes from Equation (14) in Section 4.1 and represents the truck-stall replenishment routing cost.

The formula of $l_s(x; \rho, r)$ depends on the choice of the service mode.

- In the on-demand mode, $l_s(x; \rho(x), r(x)) = \mathbb{E}[L_r(\rho(x))]\Lambda(x; \rho(x), r(x))$, where $\mathbb{E}[L_r(\rho(x))]$ is given by Proposition 3. The walking distance r is fixed at zero for the door-to-door mode. Moreover, for any given candidate decision (ρ, r) , one needs to simultaneously solve the waiting time Equation (2) and the demand Equation (16) to obtain both the endogenous demand density $\lambda(x, t; w(\rho, r), r)$ and the mean waiting time $w(\rho, r)$. The solution is unique because $w(\lambda)$ is increasing in λ , whereas $\lambda(w)$ is decreasing in w .

- In the demand-pooling mode, the stall's total traveling distance $l_s(x; \rho(x), r(x))$ is instead given by Equation (9), where $l_s = \int_{t \in T} \frac{2rN}{\mathbb{E}[T^*(t)] + N\mathbb{E}[\tau(t^p(t))]} dt$ when the arrival rates are homogenous in the local area. The average waiting time w is given by Equation (5) in Theorem 2. In our numerical studies, we replace Equation (5) with Equation (6) from Corollary 1 for ease of computation. Similar to the procedure for the on-demand mode, (5) or (6) needs to be solved jointly with the demand Equation (16) for endogenous $\lambda(x, t; w(\rho, r), r)$ and $w(\rho, r)$. The decision of the promised stay duration $t^p(x)$ also needs to be jointly optimized.

For any location $x \in \Omega$, the stall economy planning problem (SE-PP) is formulated as follows:

$$\begin{aligned} & \underset{\rho(x), r(x)}{\text{maximize}} && R(x; \rho(x), r(x)) \\ & \text{subject to} && 0 < \rho(x) \leq \rho_{\max}, \\ & && 0 \leq r(x) \leq \frac{1}{\sqrt{2\rho(x)}}, \quad \forall x \in \Omega. \end{aligned} \quad (\text{SE-PP})$$

The first constraint imposes an upper limit on stall density $\rho_{\max}$ because the municipality may limit the number of stalls the retailer can deploy on the street. The second constraint specifies that the maximum service radius is the radius of the service zone.

We solve Problem (SE-PP) numerically by grid search to obtain the optimal design $\rho^*(x)$ and $r^*(x)$ (as well as $t^p(x)$ in the demand pooling mode) at each location x . The continuous service zone design $\rho^*(x)$ can then be discretized for implementation. (See Ouyang and Daganzo (2006) for a discretization procedure, which

is beyond the scope of our paper.) Problem (SE-PP) also can be extended to represent a hybrid scenario in which the choice of the service mode varies by time, which we omit for clarity of discussion.

4.3.1. Benchmark: Stationary Stores. For benchmarking purposes, we also consider a scenario in which stalls do not move. This scenario is equivalent to setting the service radius $r(x)$ to the maximum value $1/\sqrt{\rho(x)}$ (so that all customers have to walk to the center of their service zones to visit a stall) and replacing the routing cost coefficient $\sqrt{2c_r c_s}$ with $\sqrt{2c_t}$ in Problem (SE-PP). The stall repositioning cost thus becomes zero. Solving this modified problem yields the optimal density of stationary stalls and the associated profit.

5. Computational Studies and Managerial Insights

The stall economy planning model in Section 4 allows us to evaluate whether and how retailers should run their business on wheels. We mainly examine three dimensions: the repositioning efficiency of the magnet rule (Section 5.1), the value of operational flexibilities (Section 5.2), and in a more empirically grounded and simulation-verified setting, the effects of service quality (Section 5.3). Data and source code needed to reproduce the results in this paper are included in the online appendix.

5.1. Validation of the Magnet Rule

We start with testing the repositioning efficiency of the magnet rule, which is the building block for solving the rendezvous problem in Section 3.3. Proposition 2 already establishes that the magnet rule results in the shortest stall repositioning distance to meet the next customer. Here, we further use simulation to compare the magnet rule with three alternative rules of choosing rendezvous.

- The “straight-ahead” rule. Under this rule, if the next customer's location z_c is already within the small diamond $\mathcal{I}(z_o)$ around the stall's current location z_o , then the stall stays at z_o ; otherwise, the chosen rendezvous is simply the intersection of the boundary of the small diamond $\mathcal{I}(z_c)$ around z_c and the line segment between z_o and z_c . In other words, the stall approaches the next customer as if it heads straight toward the customer's location (as opposed to being restricted to the eight directions under the magnet rule for type 1 origins). The straight-ahead rule also can be verified to guarantee the shortest stall repositioning distance to meet the next customer.

- The “two-location clairvoyant” rule and the “three-location clairvoyant” rule. We also benchmark the magnet rule against the scenarios in which the stall dispatch is optimized given the information of future customer

locations. Clearly, such clairvoyant dispatch would outperform any on-demand dispatch that only takes into account the location of the immediate next customer. Because it is difficult to solve for the optimal dispatch based on all future customer locations, we only simulate the scenarios in which the locations of the next two or three customers are given.

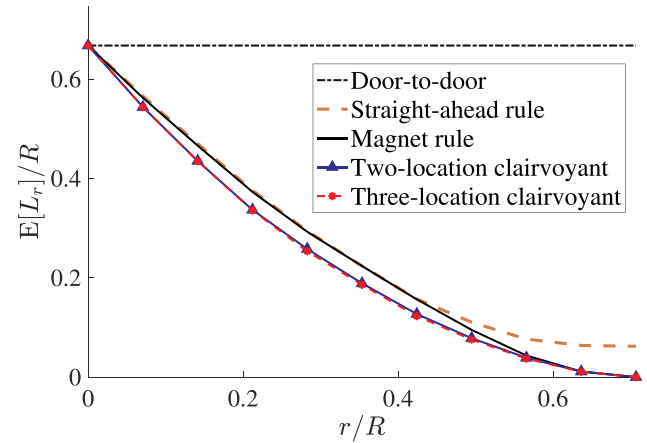
Figure 6 shows the mean stall repositioning trip distance (normalized by R , the edge length of the service zone) under different rendezvous-picking rules. Each of these mean distances is generated based on simulation of 50,000 consecutive stall moves. We draw several observations. (i) The stall repositioning distances quickly decay to zero as the customer walking radius r increases from zero to $R/\sqrt{2}$, confirming the effect of avoiding the last “100 meters” as theorized in Proposition 3. (ii) The magnet rule is more efficient than the straight-ahead rule, especially for large r . This is because, different from the magnet rule, the straight-ahead rule does not restrict the stall from exceeding the boundary of the inner diamond $\mathcal{I}_0$ (which shrinks as r increases), which is unnecessary and incurs efficiency loss in the long run. Online Appendix C provides visualization of the simulated rendezvous distributions to show this contrast. This contrast endorses the sophistication when we define the magnet rule to ensure that the stall moves only within $\mathcal{I}_0$. (iii) The magnet rule is less efficient than the two-location clairvoyant and three-location clairvoyant rules, signifying the value of knowing future customer locations. However, the gap is moderate (9.9% on average). Moreover, the fact that the curve associated with the three-location clairvoyant rule almost overlaps with that of the two-location clairvoyant rule suggests that the marginal gain of predicting into the far future is very limited. Collectively, these observations validate the magnet rule as an effective stall dispatch policy.

5.2. Mobility and Operational Flexibilities

A key question that drives this research is whether the stall economy is scalable (in terms of profitability) to meet the anticipated increasing demand for such service. Therefore, we discuss the operational flexibilities that stall mobility offers and their implications for business scalability.

5.2.1. Parameters. We set the average product profit as $p = \$10$ per customer, the distance-based stall cost as $c_s = \$2$ per kilometer, the truck cost as $c_t = \$2.73$ per kilometer, and the daily stall permit cost as $c_p = \$200$. The maximum stall deployment density is set at $\rho_{\max} = 2$ kilometers⁻². In the valuation function (15), set $\alpha_0 = 1$, $\alpha_1 = 1$, and $\alpha_2 = 3$. Other parameters include the customer shopping time at a deterministic value $\frac{1}{\mu} = 5$ minutes, the stall moving speed $v = 10$ kilometers · hour⁻¹ (here, we ignore the speed uncertainty), and the ratio of the average vacant space of a stall upon replenishment to the average

Figure 6. (Color online) Mean Repositioning Trip Distances Under Different Rendezvous-Picking Rules



volume of products purchased by a customer $S/s = 30$. (Online Appendix D provides more discussion about and justification for these parameter values.)

Without loss of generality, we consider one location and one time period and let the potential customer demand density λ_0 vary from 5 to 50 kilometers⁻² hour⁻¹ to represent different scales of market penetration. At each scale, we solve (SE-PP) for the optimal stall deployment densities ρ^* and service radius r^* under the four service modes: stationary (i.e., the benchmark), door to door, within radius, and demand pooling. Figure 7 illustrates the evolution of the optimal profits, optimal stall densities, mean waiting times, and optimal service radii as the stall economy scales up. We obtain the following findings.

5.2.2. Value of Mobility. We start by comparing the stationary and door-to-door service modes. When the potential demand is low or moderate (i.e., $\lambda_0 < 36$), Figure 7(a) shows that the gap between the profit densities in the door-to-door service mode and in the stationary mode is positive. This positive gap underscores the key advantage offered by stall mobility. To illustrate, Figure 7(b) shows that, when the demand is low and the stall density is not capped, the stationary mode entails more dense deployment of stalls to reach customers. Even so, Figure 7(d) shows that the stationary mode requires a much longer walk, thus diminishing customer demand. In contrast, the door-to-door mode allows a lower density of stall deployment so that each stall covers a larger service zone by moving around yet with little to no customer walking needed.

However, when the potential demand is moderate or large (i.e., $\lambda_0 \geq 25$), the profit of the door-to-door service mode plateaus. Too frequent repositioning not only drives up the repositioning cost but also, results

in excessive customer waiting, as Figure 7(c) shows, and customers are less likely to use the service.

5.2.3. Value of Avoiding the “Last 100 Meters.” If we switch from the door-to-door mode to the within-radius mode (i.e., avoiding traveling the last r distance to a customer’s doorstep), Figure 7(a) shows a larger profit as the potential demand scales up. As we discussed in Section 3.3, avoiding the last r distance not only reduces the stall repositioning cost but also, increases the service rate and shortens the wait, as Figure 7(c) shows. Optimizing r gives retailers an extra opportunity to strike a balance between customer waiting and walking to induce demand.

5.2.4. Value of Demand Pooling. Switching from the within-radius mode to the demand-pooling mode can further enhance profitability when the potential demand rate is high (i.e., $\lambda_0(x) > 33$). Pooling spatially and temporally adjacent demands has two effects: (i) reducing the repositioning frequency by basing it on the rate of moving across subzones rather than on the rate of demand arrivals and (ii) reducing the repositioning trip length

from $\mathbb{E}[L_r]$ on average to a deterministic distance $2r$. Consequently, stall repositioning in the pooling mode is insensitive to demand increases (which Theorem 2 already reveals), whereas the on-demand service mode has to suppress the increase in waiting time by extending the service radius r (as Figure 7(d) shows), which hurts demand.

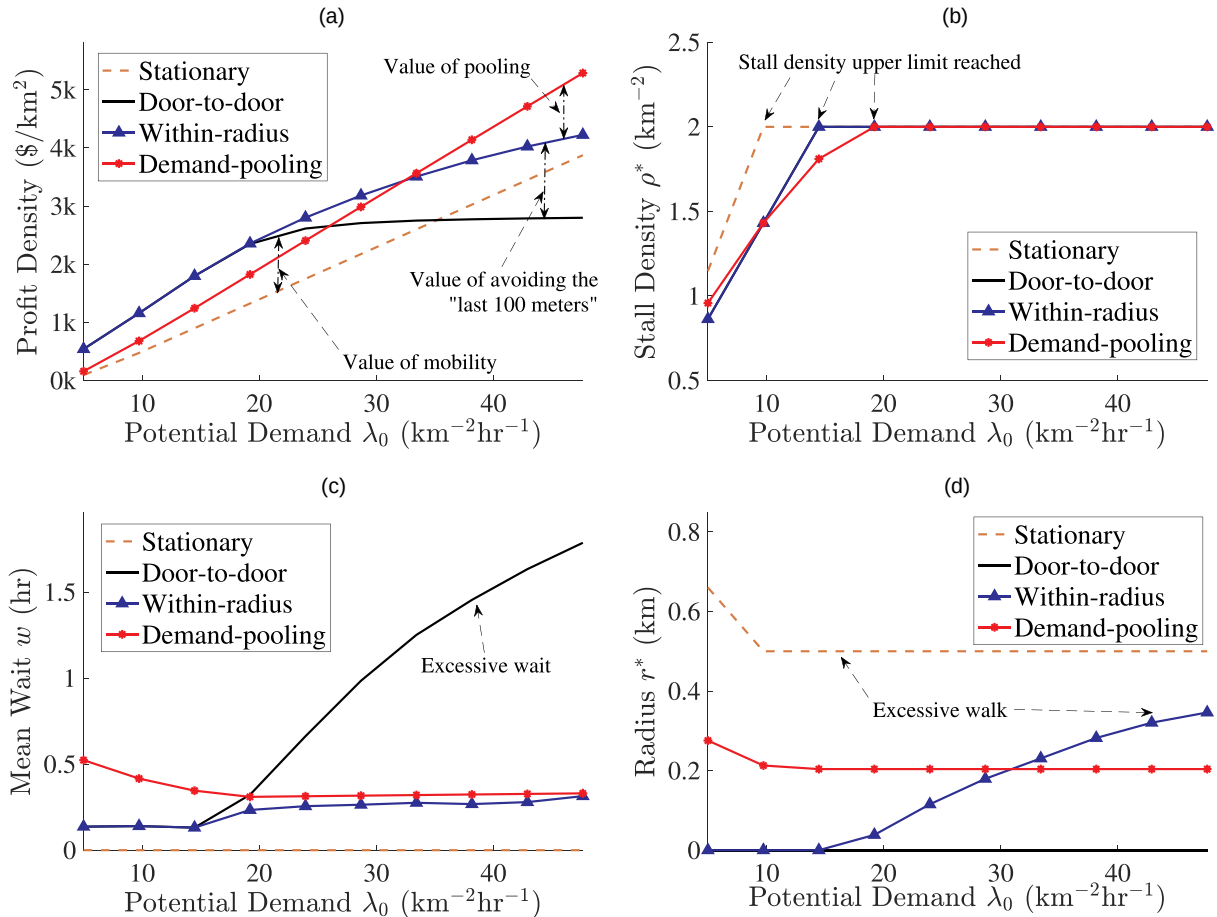
In contrast, pooling demand is less profitable when the demand is sparse. In this case, the value of pooling is insignificant. The aforementioned positive effects are dominated by the negative effects of relatively long waits and walks, as Figure 7, (c) and (d) shows.

5.2.5. Implications for Scalability. Is the stall economy scalable? The previous analysis leads to the following insights.

I. Other things being equal, the stall economy leads to more profit than stationary retail. This profitability is achieved by offering proximity to customers (as well as saving the inventory routing cost) stemming from the mobility of wheeled stalls.

II. The efficiency of the stall economy also is partly attributable to the additional *operational flexibilities* that

Figure 7. (Color online) Evolution of (a) Profit Density, (b) Optimal Stall Densities, (c) Mean Customer Wait Times, and (d) Optimal Radii as Potential Demand Scales Up



stall mobility offers. In the presence of low customer demand, stalls can function in the on-demand, within-radius (or even door-to-door) mode, swiftly repositioning to the next customer. As demand scales up, the on-demand mode can give way to the demand-pooling mode to avoid both excessive repositioning of stalls and increased waiting and walking for customers. The stall economy also may exploit additional operational flexibilities, such as customizing the service radius to each customer in the on-demand mode. We leave this extra flexibility for future study.

To summarize, our analysis demonstrates that the stall economy potentially can tap large economic opportunities because of the operational flexibilities related to stall mobility. Note, however, that our findings come from a simplified setting and do not consider other values that brick-and-mortar stores offer (e.g., larger product assortments and human assistance).

5.3. Impacts of Service Quality

Ultimately, whether the stall economy will thrive hinges on the quality of service that customers perceive or receive. Too much wait for a stall arrival, too long a walk to meet a stall, or too short a time to select products may fail to attract customers from the traditional online and offline territories. From an operational perspective, understanding how these factors pose challenges to stall operations is important. For this purpose, we examine three key parameters concerning service quality perception or reception in our model: the coefficients of the mean waiting time and the service radius in the customer valuation function (i.e., α_1 and α_2 in (15)) and the time that a customer spends shopping at a stall ($1/\mu$). In each experiment, we double one of these three values to represent a more challenging operating scenario.

We illustrate our results with an area consisting of 36 forward sortation areas (FSAs) in the city of Toronto, Canada. This urban area represents a nonhomogenous setting, including both the densely populated downtown (e.g., M5V) and less populated suburbs (e.g., M5M). Assume that the potential demand in each FSA is 5% of its 2016 population (from statcan.gc.ca). The mean, the standard deviation, the minimum, and the maximum of the potential demand density over these FSAs are 40.4, 25.3, 12.0, and 130.5 hours⁻¹ kilometer⁻², respectively. We ignore variations within each FSA or across hours. (The stall operations adapt to demand's temporal heterogeneity in a similar fashion as to demand's heterogeneity across FSAs. We focus on the latter heterogeneity.) In addition, we calibrate the stall moving speed in each FSA as the average biking speed. The average is taken over sampled biking speeds. Each sample speed is the ratio of the distance between two randomly generated locations in that FSA to the estimated travel time queried via Google Distance Matrix API. In total, we obtain 22,669 queries to account for all

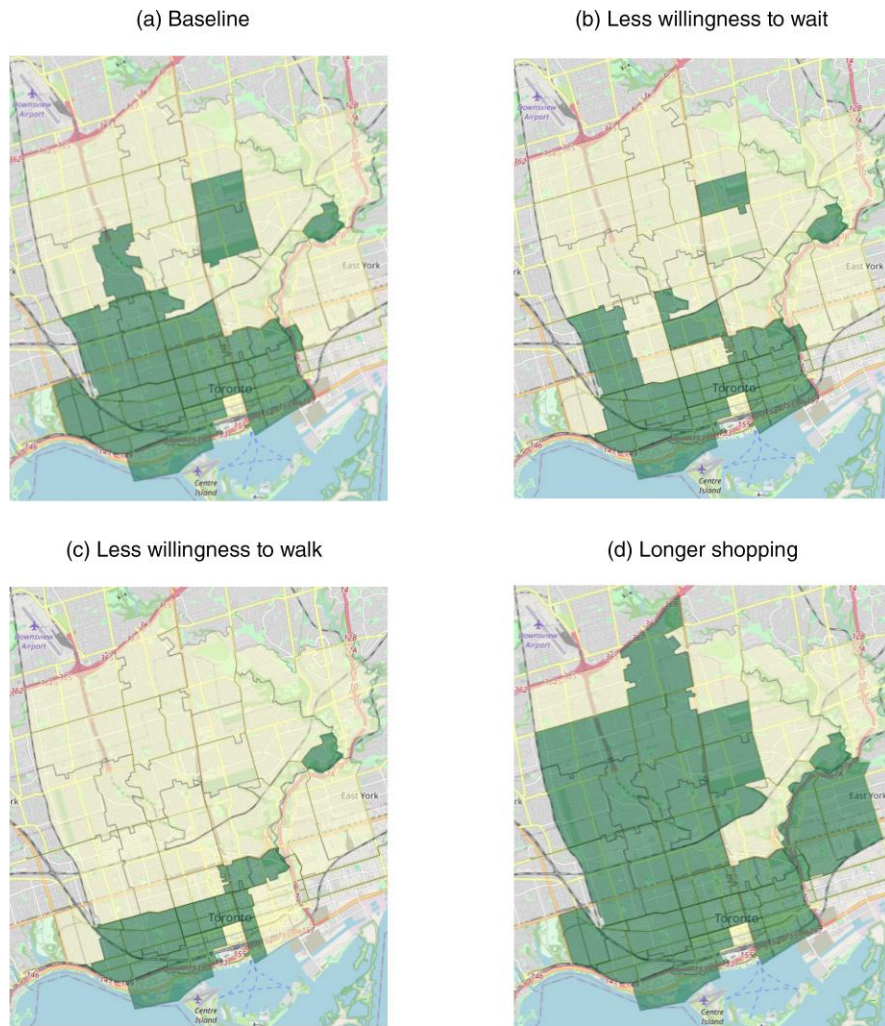
FSAs and different times of the day. The mean, the standard deviation, the minimum, and the maximum of the stall moving speed over these FSAs are 15.5, 1.80, 10.99, and 17.91 kilometers·hour⁻¹, respectively.

Figure 8, (b)–(d) visualizes the changes in the optimal service modes relative to the baseline scenario in Figure 8(a). On these maps, FSAs painted with the deeper (green) color profit more in the demand-pooling service mode, whereas the other FSAs favor the on-demand (within-radius) service mode. Our results lead to the following insights.

5.3.1. Less Willingness to Wait. If we double the coefficient of the mean waiting time, α_1 , the mean waiting time drops by 30.5% from 15.8 to 11.0 minutes; in this case, the mean service radius must increase by 17.4% from 0.14 to 0.16 kilometers. The overall profit density decreases by 10.0% on average. Comparing Figure 8(b) with Figure 8(a) shows that five more FSAs switch from the demand-pooling service mode to the on-demand mode. The reason is that when customers are less patient, the benefit of demand pooling in these FSAs becomes dominated by the demand loss because of longer waits, which are caused by demand pooling's repositioning scheme.

5.3.2. Less Willingness to Walk. Alternatively, doubling α_2 , the coefficient of the service radius, reduces profit by 17.8%. In response to this change, the service radius shrinks by 40.6% to 0.08 kilometers on average so that customers can walk less distance to meet stalls. However, customers' mean waiting time increases by 55.6%. In the demand-pooling mode, these changes occur because penalizing on the service radius forces stalls to stop at more subzones before serving the next customer in the queue, thus resulting in prolonged waits. Following a similar logic as in the previous scenario, nine FSAs have to switch to the on-demand service mode, as Figure 8(c) shows.

5.3.3. Longer Shopping. Finally, if we double $1/\mu$ so that customers spend a longer time at stalls selecting their products, the mean waiting time and the mean service radius have to rise (by 34.6% and 56.8%, respectively). The profit decreases by 11.5%. Interestingly, in contrast to the two previous scenarios involving less willingness to wait and to walk, Figure 8(d) shows that 10 more FSAs actually switch to the demand-pooling mode. The reason is that in the demand-pooling mode, when customers spend a longer time shopping, the stall has to stay longer at each subzone before moving to the next subzone. Staying a longer time allows the stall to pool more temporally and spatially adjacent demands and thus, amplifies the value of demand pooling by accommodating greater demand without significantly prolonging the wait. Only in the six FSAs

Figure 8. (Color online) FSA-Level Service Mode Choice in the Four Scenarios

Note. The deeper color indicates the FSAs where the pooling service mode is more profitable than the on-demand service mode.

with the sparsest demands does the benefit of demand pooling remain dominated by the negative effect of a prolonged wait.

This discussion again highlights the value of the operational flexibilities in the stall economy. The service mode can be readily adjusted to adapt to different service quality considerations.

5.3.4. Validation of the Spatial Queueing Models. Finally, based on this nonhomogenous urban setting, we develop simulation programs to validate the spatial queueing models of the on-demand mode and the demand pooling mode. For each combination of an FSA and a service mode, we simulate 50,000 customer order arrivals and stall moves following the optimal stall deployment density, service radius, and endogenous demand arrival rate. The mean absolute percentage error between the mean customer waiting time generated

from the simulation and that computed from the models is 5%. These numerical results validate our spatial queueing models.

6. Discussion

This paper is motivated by the observation that the recent advances in self-driving technologies point to a prospective boom in the stall economy, where retailers sell products in public spaces using wheeled vending stalls. This work aims to understand this new retail paradigm—in particular, the interdependence among customer wait and walk and stall movements. To this end, we develop spatial queueing models for serving customers using an on-demand mode and a demand-pooling mode. The on-demand mode also motivates us to propose and solve the rendezvous problem to characterize the spatial distribution of stall locations, which are state dependent and nonuniform. By studying the

optimal scheme of stall deployment in an urban setting, we find that the stall economy has potential for providing high-quality and profitable service because stall mobility provides both proximity to customers and operational flexibilities, such as avoiding the “last 100 meters” and pooling demands. In addition, we reveal how flexible stall operations can accommodate service quality enhancements.

This paper represents an early attempt to understand stall economy operations based on analytical models. More knowledge concerning this emerging retail paradigm awaits efforts to discover.

6.1. Alternate Stall Economy Service Modes

Undoubtedly, actual stall service operations should not be limited to the on-demand and demand-pooling modes considered in this paper. Between these two extremes may exist other variants. If customers need to order the stall ahead of time so that multiple subsequent customer locations are known, then the “clairvoyant” policies considered in Section 5.1 will become realistic for enhancing repositioning efficiency (by a moderate margin). This enhancement can potentially be amplified if we further relax the FCFS restriction. In that case, vehicle routing comes into play, and the stall economy operations become more akin to last-mile deliveries. The repositioning trip distance may scale with the demand λ as $\mathcal{O}(\sqrt{\lambda})$ as opposed to $\mathcal{O}(\lambda)$ in the FCFS mode or $\mathcal{O}(1)$ in the demand-pooling mode. Depending on whether to minimize the repositioning trip distance or customer wait, the underlying routing problem will be similar to the classical traveling salesman problem (TSP) or its variants, such as the traveling repairman problem (TRP) and TSP with time windows. To account for real-time customer order arrivals, the routing problem may also relate to the dynamic TRP as well as its heuristic policies (Bertsimas and Van Ryzin 1991, 1993). Nevertheless, it remains unclear how those vehicle routing contexts in the literature can be adapted to the “within-radius” setup, which we emphasize in this paper to account for customers’ wait versus walk trade-offs. The CA model of inventory replenishment developed in Section 4.1 is along this line of combining vehicle routing and customer mobility, but more analysis is worth future research.

6.2. Other Dimensions

Another direction for future research is to broaden the scope—for example, interfacing the stall economy with traditional online and offline channels, taking into account product recommendation and assortment decisions, and making use of the mobile stalls for marketing purposes. Tackling these questions can further deepen our understanding of the stall economy as we move toward a self-driving-empowered future of retail.

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References

- Ansari S, Başdere M, Li X, Ouyang Y, Smilowitz K (2018) Advancements in continuous approximation models for logistics and transportation systems: 1996–2016. *Transportation Res. Part B Methodological* 107:229–252.
- Banerjee S, Freund D, Lykouris T (2021) Pricing and optimization in shared vehicle systems: An approximation framework. *Oper. Res.* 70(3):1783–1805.
- Banerjee S, Riquelme C, Johari R (2015) Pricing in ride-share platforms: A queueing-theoretic approach. Preprint, submitted February 10, <http://dx.doi.org/10.2139/ssrn.2568258>.
- Beardwood J, Halton JH, Hammersley JM (1959) The shortest path through many points. *Math. Proc. Cambridge Philos. Soc.* 55(4): 299–327.
- Belavina E (2021) Grocery store density and food waste. *Manufacturing Service Oper. Management* 23(1):1–18.
- Benjaafar S, Ding J-Y, Kong G, Taylor T (2022a) Labor welfare in on-demand service platforms. *Manufacturing Service Oper. Management* 24(1):110–124.
- Benjaafar S, Wu S, Liu H, Gunnarsson EB (2022b) Dimensioning on-demand vehicle sharing systems. *Management Sci.* 68(2): 1218–1232.
- Bertsimas DJ, Van Ryzin G (1991) A stochastic and dynamic vehicle routing problem in the euclidean plane. *Oper. Res.* 39(4):601–615.
- Bertsimas DJ, Van Ryzin G (1993) Stochastic and dynamic vehicle routing with general demand and interarrival time distributions. *Adv. Appl. Probab.* 25(4):947–978.
- Besbes O, Castro F, Lobel I (2022) Spatial capacity planning. *Oper. Res.* 70(2):1271–1291.
- Bimpikis K, Candogan O, Saban D (2019) Spatial pricing in ride-sharing networks. *Oper. Res.* 67(3):744–769.
- Cachon GP (2014) Retail store density and the cost of greenhouse gas emissions. *Management Sci.* 60(8):1907–1925.
- Cachon GP, Daniels KM, Lobel R (2017) The role of surge pricing on a service platform with self-scheduling capacity. *Manufacturing Service Oper. Management* 19(3):368–384.
- Cao J, Olvera-Cravioto M, Shen Z-J (2020) Last-mile shared delivery: A discrete sequential packing approach. *Math. Oper. Res.* 45(4): 1466–1497.
- Carlsson JG, Song S (2018) Coordinated logistics with a truck and a drone. *Management Sci.* 64(9):4052–4069.
- Carlsson JG, Behroozi M, Mihic K (2018) Wasserstein distance and the distributionally robust TSP. *Oper. Res.* 66(6):1603–1624.
- Carlsson JG, Behroozi M, Devulapalli R, Meng X (2016) Household-level economies of scale in transportation. *Oper. Res.* 64(6): 1372–1387.
- Caro F, Kök AG, Martínez-de Albéniz V (2020) The future of retail operations. *Manufacturing Service Oper. Management* 22(1):47–58.
- Castillo JC, Knoepfle D, Weyl G (2017) Surge pricing solves the wild goose chase. *Proc. 2017 ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 241–242.
- Chen M, Hu M (2020) Courier dispatch in on-demand delivery. Preprint, submitted August 16, <http://dx.doi.org/10.2139/ssrn.3675063>.
- Chen M, Hu M, Wang J (2022) Food delivery service and restaurant: Friend or foe? *Management Sci.* 68(9):6539–6551.

- Daganzo CF (2005) *Logistics Systems Analysis* (Springer Science & Business Media, New York).
- Feldman P, Frazelle AE, Swinney R (2022) Managing relationships between restaurants and food delivery platforms: Conflict, contracts, and coordination. *Management Sci.*, ePub ahead of print March 28, <https://doi.org/10.1287/mnsc.2022.4390>.
- Feng G, Kong G, Wang Z (2021) We are on the way: Analysis of on-demand ride-hailing systems. *Manufacturing Service Oper. Management* 23(5):1237–1256.
- Gallino S, Moreno A (2019) *Operations in an Omnichannel World* (Springer, Berlin).
- Glaeser CK, Fisher M, Su X (2019) Optimal retail location: Empirical methodology and application to practice. *Manufacturing Service Oper. Management* 21(1):86–102.
- Global Times (2021) China delivery giant Meituan raises \$10b. (April 22), <https://www.globaltimes.cn/page/202104/1221797.shtml>.
- Gurvich I, Lariviere M, Moreno A (2019) Operations in the on-demand economy: Staffing services with self-scheduling capacity. Hu M, ed. *Sharing Economy* (Springer, Berlin), 249–278.
- Lim MK, Mak H-Y, Shen Z-JM (2017) Agility and proximity considerations in supply chain design. *Management Sci.* 63(4):1026–1041.
- Liu S, He L, Shen Z-JM (2021) On-time last-mile delivery: Order assignment with travel-time predictors. *Management Sci.* 67(7): 4095–4119.
- Mao W, Ming L, Rong Y, Tang CS, Zheng H (2019) Faster deliveries and smarter order assignments for an on-demand meal delivery platform. Preprint, submitted October 12, <http://dx.doi.org/10.2139/ssrn.3469015>.
- Newell GF (1971) Dispatching policies for a transportation route. *Transportation Sci.* 5(1):91–105.
- Ouyang Y, Daganzo CF (2006) Discretization and validation of the continuum approximation scheme for terminal system design. *Transportation Sci.* 40(1):89–98.
- Qi W, Li L, Liu S, Shen Z-JM (2018) Shared mobility for last-mile delivery: Design, operational prescriptions, and environmental impact. *Manufacturing Service Oper. Management* 20(4): 737–751.
- Reed S, Campbell AM, Thomas BW (2022) The value of autonomous vehicles for last-mile deliveries in urban environments. *Management Sci.* 58(1):280–299.
- Shortle JF, Thompson JM, Gross D, Harris CM (2018) *Fundamentals of Queueing Theory*, vol. 399 (John Wiley & Sons, New York).
- Stop & Shop (2019) Stop & Shop brings on-demand, self-driving grocery stores to consumers driverless vehicles set to launch in greater Boston in 2019. Accessed May 12, 2021, <https://stopandshop.com/pages/stsh-bring-on-demand-self-driving-grocery-stores-to-consumers>.
- Taylor TA (2018) On-demand service platforms. *Manufacturing Service Oper. Management* 20(4):704–720.
- TechSpot (2019) Stop & Shop's pilot program brings the grocery store to your door. Accessed August 31, 2020, <https://www.techspot.com/news/78312-stop-shop-pilot-program-brings-grocery-store-door.html>.
- Walmart (2021) Walmart invests in Cruise, the all-electric self-driving company. Accessed May 13, 2021, <https://corporate.walmart.com/newsroom/2021/04/15/walmart-invests-in-cruise-the-all-electric-self-driving-company>.
- Yildiz B, Savelsbergh M (2019) Service and capacity planning in crowd-sourced delivery. *Transportation Res. Part C Emerging Tech.* 100:177–199.

Junyu Cao is an assistant professor in the Department of Information, Risk, and Operations Management at the University of Texas at Austin. Her research interests include data-driven decision making and stochastic modeling, with applications to smart-city operations and revenue management.

Wei Qi is an associate professor in the Department of Industrial Engineering of Tsinghua University. His research interests center on smart-city operations concerning urban energy, mobility, retail logistics, and climate adaptation.